

PracticeMidterm1_VA (2)

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Math 340: Elementary Matrix and Linear Algebra

PRACTICE MIDTERM ONE

Name: _____	Section#: _____	NetID: _____
I pledge that the work on this exam is entirely my own. I understand that the penalties for cheating may include an F in the course and referral to my dean for further action.		
Student Signature: _____		

READ THE FOLLOWING INFORMATION.

- This is a 90-minute exam. It consists of five problems on five pages; the exam is 4 sheets of paper, including this cover sheet. It is your responsibility to make sure that you have a complete exam.
- Books, notes, and other aids are not allowed.
- You may use the backs of the pages or any additional pages for scratch work, but it will not be graded. Do not unstaple or remove pages as they can be lost in the grading process.
- Only complete, well written, and neat solutions will be awarded full credit. Remember that all claims must be supported. Responses which do not meet these qualifications may be awarded some partial credit.

DO NOT BEGIN THIS EXAM UNTIL SIGNALLED TO DO SO.

1. (2+4+4 points) Let

$$A = \begin{bmatrix} 3 & -1 \\ 2 & 7 \\ 4 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} -2 & 0 \\ 1 & 4 \\ 5 & -3 \end{bmatrix}, \quad \text{and } C = \begin{bmatrix} 4 & -6 \\ 0 & 2 \end{bmatrix}$$

be three matrices. Compute the following (if possible). If the computation is not possible, explain why not.

$$(a) \quad 3B - A = \begin{bmatrix} -6 & 0 \\ 3 & 12 \\ 15 & -9 \end{bmatrix} - \begin{bmatrix} 3 & -1 \\ 2 & 7 \\ 4 & 0 \end{bmatrix} = \begin{bmatrix} -9 & 1 \\ 1 & 5 \\ 11 & -9 \end{bmatrix}$$

$$(b) \quad B^T A - C = \begin{matrix} 2 \times 3 \\ \begin{bmatrix} -2 & 1 & 5 \\ 0 & 4 & -3 \end{bmatrix} \end{matrix} \begin{matrix} 3 \times 2 \\ \begin{bmatrix} 3 & -1 \\ 2 & 7 \\ 4 & 0 \end{bmatrix} \end{matrix} - \begin{bmatrix} 4 & -6 \\ 0 & 2 \end{bmatrix}$$

fixed mistake

$$= \begin{bmatrix} 16 & 9 \\ -4 & 28 \end{bmatrix} - \begin{bmatrix} 4 & -6 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 12 & 15 \\ -4 & 26 \end{bmatrix}$$

(c) $C^T B$
 $\hookrightarrow 2 \times 2 \quad 3 \times 2$
 $\quad \quad \quad \times$ can't multiply because their inner dimensions do not match.

2. (4+3+4 points) Let

$$D = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 2 & 0 \\ 0 & 1 & 1 \end{bmatrix}, \quad G = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 2 & 0 \\ 3 & 7 & 1 \end{bmatrix}, \quad \text{and } H = \begin{bmatrix} -4 & 0 & -4 \\ 1 & 2 & 0 \\ 0 & 1 & 1 \end{bmatrix}$$

be three matrices.

(a) Find matrices E and F such that $ED = G$ and $FD = H$.

Note:

D and G only differ in r_3 . In fact: $D \xrightarrow{3r_2+r_3} G$

D and H differ in r_1 : $D \xrightarrow{-4r_1+r_2} H$

elementary matrix
 $E = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 3 & 1 \end{bmatrix}$

$$F = \begin{bmatrix} -4 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

(b) Find E^{-1} .

undo E by taking $-3r_2 + r_3$.

$$E^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -3 & 1 \end{bmatrix}$$

$$(c) \text{ Find } (EF)^{-1} = F^{-1}E^{-1} = \begin{bmatrix} -1/4 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -3 & 1 \end{bmatrix} = \begin{bmatrix} -1/4 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -3 & 1 \end{bmatrix}$$

3. (3+3+4 points) Let

$$K = \begin{bmatrix} 1 & 0 & 1 \\ a_1^2 - a_1 & 2 & a_2^2 \\ 1 & 9 & 1 \end{bmatrix}, \quad L = \begin{bmatrix} 0 & b_1 & 2 \\ 1 & 0 & 3 \\ -2 & b_2 & 0 \end{bmatrix}, \quad \text{and } M = \begin{bmatrix} 2 & 1 \\ 0 & 0 \\ 6 & 3 \end{bmatrix}$$

be three matrices.

(a) Find all pairs (a_1, a_2) which make K symmetric. $\rightarrow K^T = K$:

$$K^T = \begin{bmatrix} 1 & a_1^2 - a_1 & 1 \\ 0 & 2 & a_2^2 \\ 1 & a_2^2 & 1 \end{bmatrix} \stackrel{?}{=} \begin{bmatrix} 1 & 0 & 1 \\ a_1^2 - a_1 & 2 & a_2^2 \\ 1 & a_2^2 & 1 \end{bmatrix}$$

Need: $0 = a_1^2 - a_1 = a_1(a_1 - 1)$
 $a_1 = 0, 1$

$9 = a_2^2 \rightarrow a_2 = \pm 3$

$$\boxed{\begin{matrix} (0, 3) & (0, -3) \\ (1, 3) & (1, -3) \end{matrix}}$$

(b) Find $b_1 + 4b_2$ if L is skew symmetric. $\rightarrow L^T = -L$:

$$L^T = \begin{bmatrix} 0 & 1 & -2 \\ b_1 & 0 & b_2 \\ 2 & 3 & 0 \end{bmatrix} \stackrel{?}{=} \begin{bmatrix} 0 & -b_1 & -2 \\ -1 & 0 & -3 \\ 2 & -b_2 & 0 \end{bmatrix}$$

$-b_1 = 1 \rightarrow b_1 = -1$
 $-b_2 = 3 \rightarrow b_2 = -3$

$b_1 + 4b_2 = -1 - 12 = \boxed{-13}$

(c) Let f be a function defined by $f(\mathbf{x}) = M\mathbf{x}$. What is the domain, the codomain, and the range of f ?

$$\begin{bmatrix} 2 & 1 \\ 0 & 0 \\ 6 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 2x_1 + x_2 \\ 0 \\ 6x_1 + 3x_2 \end{bmatrix} = \begin{bmatrix} 2x_1 + x_2 \\ 0 \\ 3(2x_1 + x_2) \end{bmatrix} = \begin{bmatrix} a \\ 0 \\ 3a \end{bmatrix}$$

Domain: 2-vectors, $\mathbb{R}^2$

Codomain: 3-vectors, $\mathbb{R}^3$

Range: vectors of the form $\begin{bmatrix} a \\ 0 \\ 3a \end{bmatrix} a \in \mathbb{R}$.

4. (4+4+3 points) Let

$$N = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 3 & 4 & 0 \end{bmatrix} \text{ and } P = \begin{bmatrix} 1 & 4 & 3 & 5 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 2 & -8 \end{bmatrix}$$

be two matrices.

(a) Find N^{-1} , if it exists.

$$\begin{aligned} & \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & 4 & 0 & 1 & 0 \\ 3 & 4 & 0 & 0 & 0 & 1 \end{array} \right] \xrightarrow{-3r_1+r_3} \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & 4 & 0 & 1 & 0 \\ 0 & -2 & -9 & -3 & 0 & 1 \end{array} \right] \xrightarrow{2r_2+r_3} \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & 4 & 0 & 1 & 0 \\ 0 & 0 & -1 & -3 & 2 & 1 \end{array} \right] \\ & \xrightarrow{-r_3 \rightarrow r_2} \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & 0 & -12 & 9 & 4 \\ 0 & 0 & -1 & 3 & -2 & -1 \end{array} \right] \xrightarrow{-4r_2+r_3} \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & 0 & -12 & 9 & 4 \\ 0 & 0 & 1 & 3 & -2 & -1 \end{array} \right] \xrightarrow{-2r_2+r_1} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 16 & -12 & -5 \\ 0 & 1 & 0 & -12 & 9 & 4 \\ 0 & 0 & 1 & 3 & -2 & -1 \end{array} \right] \\ & N^{-1} = \begin{bmatrix} 16 & -12 & -5 \\ -12 & 9 & 4 \\ 3 & -2 & -1 \end{bmatrix} \end{aligned}$$

(b) Compute the reduced row echelon form of P .

$$\begin{aligned} & \left[\begin{array}{cccc} 1 & 4 & 3 & 5 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 2 & -8 \end{array} \right] \xrightarrow{R_2 \leftrightarrow R_4} \left[\begin{array}{cccc} 1 & 4 & 3 & 5 \\ 0 & 0 & 2 & -8 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{array} \right] \xrightarrow{\frac{1}{2}R_2 \rightarrow R_2, \frac{1}{3}R_3 \rightarrow R_3} \left[\begin{array}{cccc} 1 & 4 & 3 & 5 \\ 0 & 0 & 1 & -4 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{array} \right] \\ & \xrightarrow{4R_2+R_1 \rightarrow R_1, -5R_2+R_1 \rightarrow R_1} \left[\begin{array}{cccc} 1 & 4 & 3 & 0 \\ 0 & 0 & 1 & -4 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{array} \right] \xrightarrow{-3R_2+R_1 \rightarrow R_1} \left[\begin{array}{cccc} 1 & 4 & 0 & 0 \\ 0 & 0 & 1 & -4 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{array} \right] \end{aligned}$$

(c) Identify all solutions to the linear system $Px = 0$.

soln set: $\begin{bmatrix} -4t \\ t \\ 0 \\ 0 \end{bmatrix}, t \in \mathbb{R}.$

use this: x_2 is a free variable.

$$x_2 = t$$

$$x_4 = 0$$

$$x_3 = 0$$

$$x_1 + 4x_2 = 0 \Rightarrow x_1 = -4t$$

5. (2+6 points) Solve the following linear system

$$\begin{array}{rclcrcl} 3x & - & 5y & + & 2z & = & 3 \\ -2x & + & 4y & + & 2z & = & -6 \\ 4x & - & 7y & + & z & = & 5 \end{array}$$

(a) Express this linear system in the form $A\mathbf{x} = \mathbf{b}$.

$$\begin{bmatrix} 3 & -5 & 2 \\ -2 & 4 & 2 \\ 4 & -7 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3 \\ -6 \\ 5 \end{bmatrix}$$

(b) If the system has a unique solution, provide it. If it has infinitely many solutions, provide an explicit parametrized form for these solutions. If it has no solutions, state this clearly. You must use Gaussian elimination or Gauss–Jordan reduction.

$$\left[\begin{array}{ccc|c} 3 & -5 & 2 & 3 \\ -2 & 4 & 2 & -6 \\ 4 & -7 & 1 & 5 \end{array} \right] \xrightarrow[R_1 \leftrightarrow R_2]{-\frac{1}{3}R_1 \rightarrow R_1} \left[\begin{array}{ccc|c} -2 & 4 & 2 & -6 \\ 3 & -5 & 2 & 3 \\ 4 & -7 & 1 & 5 \end{array} \right]$$

$$\left[\begin{array}{ccc|c} 1 & -2 & -1 & 3 \\ 3 & -5 & 2 & 3 \\ 4 & -7 & 1 & 5 \end{array} \right] \xrightarrow{\substack{-3R_1 + R_2 \rightarrow R_2 \\ -4R_1 + R_3 \rightarrow R_3}} \left[\begin{array}{ccc|c} 1 & -2 & -1 & 3 \\ 0 & 1 & 5 & -6 \\ 4 & -7 & 1 & 5 \end{array} \right] \xrightarrow{-4R_2 + R_3 \rightarrow R_3} \left[\begin{array}{ccc|c} 1 & -2 & -1 & 3 \\ 0 & 1 & 5 & -6 \\ 0 & 1 & 5 & -7 \end{array} \right]$$

$$\left[\begin{array}{ccc|c} 1 & -2 & -1 & 3 \\ 0 & 1 & 5 & -6 \\ 0 & 0 & 0 & -1 \end{array} \right] \rightarrow \text{this says: } 0 = -1. \text{ —}$$

This means: the system has no solutions
(or is inconsistent)

6. (3+3+4 points) Let

$$A = \begin{bmatrix} 5 & -a \\ a-2 & -4 \end{bmatrix}, \quad B = \begin{bmatrix} -2 & 0 & 0 & 0 \\ 3 & 4 & 0 & 0 \\ 8 & -9 & 1 & 0 \\ 6 & -7 & 3 & -3 \end{bmatrix}$$

be two matrices.

(a) Find a if $\det(A) = 15$.

$$\begin{aligned} \det(A) &= 5(-4) + a(a-2) = 15 \\ -20 + a^2 - 2a &= 15 \\ a^2 - 2a - 35 &= 0 \\ (a-7)(a+5) &= 0 \\ \boxed{a = 7, -5} \end{aligned}$$

(b) Compute $\det(B)$. This is triangular, so

$$\det(B) = (-2)(4)(1)(-3) = \boxed{24}$$

(c) The matrix C is obtained from B by the following row operations:

- (1) interchange rows 1 and 3, $-\det(B)$
- (2) multiply row 4 by $\frac{1}{6}$, $-\frac{1}{6}\det(B)$
- (3) add 2 times row 3 to row 1, $\rightarrow$ same, $-\frac{1}{6}\det(B)$
- (4) multiply row 1 by -2 , $\frac{2}{6}\det(B) = \frac{1}{3}\det(B)$
- (5) subtract 5 times row 2 from row 4.

What is $\det(C)$?

$$\hookrightarrow \text{same: } \frac{1}{3}\det(B) = \frac{1}{3} \cdot 24 = \boxed{8}$$

7. (4+4+5 points) Let

$$\begin{matrix} \nearrow R_2 + R_1 \rightarrow R_1 \\ \left[\begin{array}{ccc} 0 & -5 & 1 \\ 0 & -2 & 2 \\ 1 & 3 & -1 \end{array} \right] \text{ save det.} \end{matrix}$$

$$D = \begin{bmatrix} 2 & 1 & -1 \\ 0 & -2 & 2 \\ 1 & 3 & -1 \end{bmatrix}, \quad G = \begin{bmatrix} 1 & 0 & 1 \\ 1 & b & 2 \\ 0 & 1 & 1 \end{bmatrix}, \quad \text{and } H = \begin{bmatrix} 2 & c & 5 \\ 0 & 1 & 4 \\ 8 & -2 & 0 \end{bmatrix}$$

be three matrices.

(a) Find b if $\det(DG) = 40$.

$$\det(D) = 1 \cdot (-5 \cdot 2 - (-2)(1)) = -8 \Rightarrow (-8) \det(G) = 40$$

$$\det(G) = -5$$

$$\det(G) = 1 \cdot (b \cdot 1 - 2) - 0 + 1(1 - 0)$$

$$= b - 1 = -5$$

$$\boxed{b = -4}$$

(b) Let $\mathbf{b} = \begin{bmatrix} 13 \\ 10 \\ 2 \end{bmatrix}$. In one of the solutions of the linear system $H\mathbf{x} = \mathbf{b}$, $x_1 = 0$, and in another solution, $x_1 = -3$. Find c . H does not have a unique solution $\Rightarrow \det(H) = 0$.

$$\text{so } 0 = \det(H) = 2(0 + 8) - 0 + 8(4c - 5)$$

$$0 = 16 + 32c - 40 = 32c - 24$$

$\Downarrow$

$$\boxed{c = \frac{24}{32} = \frac{3}{4}}$$

(c) ~~Compute the determinant of $\text{adj } D$.~~